

Compressing Neural Training Dynamics with Response Memory

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Abstract

Compressing a trained network at one time does not by itself compress its training dynamics: two functionally identical states can respond differently to the next gradient update, and small local errors can accumulate through feedback. We study this problem for fixed-depth, fully connected nonlinear networks under feature-learning gradient-flow scaling. Every learned hidden matrix is the initialized matrix plus a time integral of outer products of a forward response and a backward response. We replace the growing response history by a fixed number P of evolving shifted-Legendre moments per neuron, sample, and internal layer. The moments form an autonomous closure, while the initialized matrices and their transposes are retained exactly.

For arbitrary fixed width, dataset, depth, initialization, and finite time horizon, we prove uniform tracking of dense gradient flow. A residual-activity clock gives physical-state error $O(P^{-1})$ for locally $C^{1,1}$ activations. A response-arclength clock controls the joint forward/backward path and gives $O(P^{-2})$ for the corresponding smoother activation class. These are global finite-horizon results: the horizon is arbitrary but the constants are not uniform over infinite training time or width. The closure compresses the learned moving correction to $O(nmP)$ coordinates per internal layer and rank at most mP (or $m(P+1)$ for the weighted clock), but it retains $O(n^2)$ initialized operators and their dense actions. It is therefore a compression of learned moving state, not yet a compression of the whole model.

Finite experiments support the construction. On five sparse-label circle tasks at width 2048, order seven reproduces the dense fitted function with RMS discrepancies from 9.4×10^{-6} to 4.7×10^{-3} . On a width-4096 MNIST 3-versus-8 experiment with 100 training images, orders one through three agree with dense predictions on 1984 held-out images to RMS 3.25×10^{-3} , 1.08×10^{-3} , and 1.20×10^{-3} . Broader tests also expose low-order failures and show that the sharper clock is not uniformly better at practical orders. We give exact complexity accounting, state the population-limit interpretation that is currently justified, and separate it from the still-open width-uniform compression problem.

1 Why training compression is different

The primary question of this paper is simple: *how much can one compress the training dynamics of a neural network?* The analogous question for a single trained snapshot is much easier to pose. One may prune or factor its weights, distill its predictions, or approximate its input–output map. None of these operations alone supplies a compressed state for continued training. Two parameter states can represent the same function on every input and still have different tangent features. Their next gradient updates can therefore differ. A small one-step discrepancy can then feed back into later activations, gradients, and updates.

We call a compression of training *global on finite horizons* if, for each finite T and tolerance ε , one can choose a finite compression order so that one autonomous compressed trajectory stays within ε of the dense trajectory for every $0 \leq t \leq T$. This is stronger than approximating the vector field at one state and weaker than a single error bound uniform over $0 \leq t < \infty$. The

state dimension should remain constant as elapsed training time grows. It may still depend on the requested accuracy, dataset size, depth, and the representation class.

This question is scientifically useful even before it produces a smaller implementation. An accurate autonomous compression identifies information that is sufficient for future learning, not merely for current prediction. It can expose which aspects of the dense trajectory carry memory and which details may be discarded. Our nonnegotiable target retains three ingredients together: more than one hidden layer, nonlinear responses that remain active, and feature learning rather than a fixed-kernel linearization.

Wide-network limits offer a natural setting for this program. With an appropriate maximal-update or mean-field scaling, nontrivial feature motion survives at large width [Yang and Hu, 2021, Nguyen and Pham, 2023]. Empirical averages may converge to deterministic fields, much as particle systems lead to continuum descriptions. Existing tensor-program, mean-field, and dynamical-field theories provide powerful exact descriptions of such limits [Yang and Hu, 2021, Nguyen and Pham, 2023, Bordelon and Pehlevan, 2022]. In the deep feature-learning settings closest to ours, however, exact descriptions retain operators, two-time kernels, or a growing family of historical responses. They do not by themselves show that the learned dynamics can be approximated by a fixed amount of history state throughout training.

The observation behind our construction is elementary. For every internal layer, gradient flow writes the learned matrix increment as an integral of rank-one products,

$$W_\ell(t) - W_{\ell,0} = -\frac{2}{nm} \sum_{a=1}^m \int_0^t r_a(s) \delta_{\ell,a}(s) h_{\ell-1,a}(s)^\top ds. \quad (1)$$

Exact history storage grows with time. We instead project the forward and backward response paths onto the first P shifted Legendre modes and evolve their coefficients online. The closure reconstructs the learned interaction from these moment pairs. Shifted Legendre memory is closely related to the online polynomial-projection principle developed by HiPPO [Gu et al., 2020]. The contribution here is to couple such memory to the forward/backward feedback of deep gradient flow, preserve the reused initialized operator in both orientations, and prove trajectory tracking for the resulting autonomous nonlinear closure.

Two clocks produce two guarantees. The first advances at residual RMS. It removes the residual factor from forward response speed and gives an $O(P^{-1})$ state bound at arbitrary fixed depth. The second advances at residual activity plus the arclength of all stored forward and backward responses. It also keeps the original learning measure inside the history integral. Both histories then have first-order projection error, and their outer-product error is $O(P^{-2})$.

The main contributions are as follows.

1. We give explicit autonomous response-memory ODEs for a fixed number of moments per neuron, sample, and internal layer. They retain each initialized matrix $W_{\ell,0}$ and its transpose exactly and never use a dense reference trajectory.
2. For arbitrary fixed width, finite dataset, fixed depth, arbitrary finite initialization, and any prescribed finite horizon, we prove uniform dense-trajectory error $O(P^{-1})$ for the activity clock and $O(P^{-2})$ for the response clock. The proof controls the closure’s own nonlinear feedback by an exact accumulator identity and a first-exit argument.
3. We distinguish three representation classes—scalar, population, and operator dynamics—and record a restricted exact scalar nonclosure theorem. The response closure compresses the learned state to population fields but remains a hybrid with fixed initialized operators.

4. We give complete state, rank, and per-step cost accounting. The learned correction has $O(nmP)$ moving coordinates per internal layer, while the exact initialized operators still require $O(n^2)$ storage and dense actions. No total-model or generic runtime improvement is claimed.
5. We test the construction on sparse circle tasks, deeper networks, several activations, sphere tasks, and MNIST. The experiments show high fidelity at small order in many cases, together with order reversals, numerical sensitivity, and difficult cases where low order fails.

The distinction between proved and prospective population claims matters. The central approximation theorem in this paper is pathwise at fixed finite width. Combining it with a separately established population limit gives a diagonal sequence of closure orders converging to the same population observables. We do not prove that one fixed order works uniformly in width, or that the closure constants remain controlled for all training time.

2 Problem, model, and representation classes

2.1 Three meanings of a finite dynamical description

The word *finite* hides substantial differences. We use the following classification.

Definition 1 (Scalar, population, and operator state). A *scalar ODE* has state in $\mathbb{R}^q$ for fixed q . A *population ODE* has a fixed finite list of scalar or vector fields on fixed probability spaces; an n -particle discretization stores one value per particle and field. An *operator ODE* has a fixed finite list of operators between population spaces. Such an operator can retain arbitrary row-column correlations and is generally not specified by finitely many population fields or scalars.

Counting only the number of named objects would call all three descriptions finite, but their information content is different. A general limiting operator is the continuum analogue of a dense matrix. Likewise, one unrestricted field can encode an arbitrary amount of information in its spatial profile. Any compression statement must therefore name both the number and the type of retained objects.

Our closure is a finite collection of neuron-population fields indexed by sample, layer, and history mode, together with fixed initialized operators. The learned moving part is population-valued; the total representation is a population/operator hybrid. At finite width these fields are simply neuron vectors.

Exact finite scalar closure is already obstructed in a broad natural class. The following result is included to calibrate the target, not to claim an impossibility theorem for every conceivable encoder.

Proposition 2 (Restricted exact scalar nonclosure). *Consider a state-universal deep linear feature-learning system with several trainable factors. Suppose an encoder consists of a fixed finite list of complete tensor contractions whose connected graph degree is bounded independently of width, its dynamics are a width-independent polynomial ODE, and the output is a polynomial readout. No such encoder exactly realizes the output dynamics on a nonempty open set for all sufficiently large widths. The same conclusion holds for analytic finite-jet closures near the zero state. It does not rule out arbitrary encoders or a closure restricted to one initialized trajectory.*

The proof, given in Appendix A.1, repeatedly differentiates the output. It produces connected contractions of unbounded graph degree, whereas polynomial operations on a fixed bounded-degree alphabet can only form disjoint unions of its existing connected components. Stable graph independence at large width makes the contradiction algebraic. This result motivates approximate population memory; it does not prove that every exact scalar compression is impossible.

2.2 Deep nonlinear feature-learning gradient flow

Fix input dimension d , sample count m , width n , and $L \geq 2$ hidden layers. The data are $(x_a, y_a)_{a=1}^m$ and $u_a = x_a/\sqrt{d}$. Hidden layers have common width only to simplify notation. With coordinatewise activations ϕ_ℓ ,

$$\begin{aligned} z_{1,a} &= W_1 u_a, & h_{1,a} &= \phi_1(z_{1,a}), \\ z_{\ell,a} &= W_\ell h_{\ell-1,a}, & h_{\ell,a} &= \phi_\ell(z_{\ell,a}), \quad 2 \leq \ell \leq L, \\ f_a &= \frac{1}{n} w^\top h_{L,a}, & r_a &= f_a - y_a. \end{aligned} \quad (2)$$

Here $W_1 \in \mathbb{R}^{n \times d}$, $W_\ell \in \mathbb{R}^{n \times n}$ for $2 \leq \ell \leq L$, and $w \in \mathbb{R}^n$. We use unhalved mean squared loss

$$\mathcal{L} = \rho^2, \quad \rho = \left(\frac{1}{m} \sum_{a=1}^m r_a^2 \right)^{1/2}. \quad (3)$$

Backpropagated response fields omit the loss derivative:

$$\begin{aligned} \delta_{L,a} &= w \odot \phi'_L(z_{L,a}), \\ \delta_{\ell,a} &= \phi'_\ell(z_{\ell,a}) \odot W_{\ell+1}^\top \delta_{\ell+1,a}, \quad \ell < L. \end{aligned} \quad (4)$$

The block mobilities $(n, 1, \dots, 1, n)$ are the canonical feature-learning scaling used throughout the paper. The resulting gradient flow is

$$\begin{aligned} \dot{W}_1 &= -\frac{2}{m} \sum_a r_a \delta_{1,a} u_a^\top, \\ \dot{W}_\ell &= -\frac{2}{nm} \sum_a r_a \delta_{\ell,a} h_{\ell-1,a}^\top, \quad 2 \leq \ell \leq L, \\ \dot{w} &= -\frac{2}{m} \sum_a r_a h_{L,a}. \end{aligned} \quad (5)$$

All layers move at order one under the intended wide scaling, so the model is not reduced to a frozen neural tangent kernel [Jacot et al., 2018, Yang and Hu, 2021].

Let $\theta = (W_1, \dots, W_L, w)$ and denote the right-hand side of (5) by $F(\theta)$. Integrating an internal-layer equation gives (1). Define, when $\rho > 0$,

$$b_{\ell,a} = \frac{r_a \delta_{\ell,a}}{\rho}. \quad (6)$$

Then the learned increment is the integral of $b_{\ell,a} h_{\ell-1,a}^\top$ against activity measure $\rho(t) dt$. The closure will compress exactly these paired histories.

2.3 Why the population regime remains the target

Under Gaussian or sub-Gaussian initialization, fixed depth and data, and globally Lipschitz activation derivatives, established multilayer mean-field theorems construct nonlinear population flows on a common local interval and identify finite-network observables as width grows [Nguyen and Pham, 2023]. Maximal-update tensor programs likewise identify feature-learning limits for discrete training [Yang and Hu, 2021]. These results justify studying a deterministic population target rather than only one finite random network.

The approximation theorem below deliberately proves something logically different: for each realized finite network it controls the response closure against that network’s dense gradient flow, for every prescribed finite horizon. No random initialization is required. This pathwise statement is useful for implementation and is a prerequisite for, but not a substitute for, a width-uniform population theorem. Section 4.3 states the exact diagonal consequence currently available.

3 Autonomous response-memory closures

3.1 Shifted Legendre coordinates

Let p_k be the shifted Legendre polynomial of degree k on $[0, 1]$, normalized by

$$p_k(1) = 1, \text{quad} \int_0^1 p_j(x)p_k(x) \, dx = \frac{\mathbf{1}_{j=k}}{2k+1}. \quad (7)$$

Their dilation identity is

$$xp'_k(x) = kp_k(x) + \sum_{j < k} (2j+1)p_j(x). \quad (8)$$

For order P , let $p = (p_0, \dots, p_{P-1})^\top$, let $e = (1, \dots, 1)^\top \in \mathbb{R}^P$, and define the lower-triangular matrix

$$A_{kk} = k, \text{quad} A_{kj} = 2j+1 \ (j < k), \text{quad} A_{kj} = 0 \ (j > k), \quad (9)$$

with indices starting at zero. Then $xp'(x) = Ap(x)$.

The same identity that makes online Legendre projection possible in signal memory [Gu et al., 2020] makes our neural closure autonomous. When the history interval expands, differentiating a moment only mixes it with moments of the same or lower degree. No stored past samples are needed.

3.2 Variant A: residual-activity clock

Set

$$\dot{\tau} = \rho, \text{quad} \tau(0) = 1. \quad (10)$$

The unit prefix $0 \leq \xi \leq 1$ carries constant forward history $h_{\ell-1,a}(0)$ and zero backward history. Physical training inserts the current pair $(h_{\ell-1,a}, b_{\ell,a})$ at $\xi = \tau(t)$. Since $d\xi = \rho dt$, the learned integral in (1) is unchanged.

For every internal link $2 \leq \ell \leq L$ and sample a , store $n \times P$ moment matrices $H_{\ell,a}$ and $U_{\ell,a}$ whose k th columns are

$$\begin{aligned} H_{\ell,a, \cdot, k} &= \int_0^\tau \bar{h}_{\ell-1,a}(\xi) p_k(\xi/\tau) \, d\xi, \\ U_{\ell,a, \cdot, k} &= \int_0^\tau \bar{b}_{\ell,a}(\xi) p_k(\xi/\tau) \, d\xi. \end{aligned} \quad (11)$$

Differentiation and (8) give the closed ODE

$$\begin{aligned} \dot{H}_{\ell,a} &= \rho h_{\ell-1,a} e^\top - \frac{\rho}{\tau} H_{\ell,a} A^\top, \\ \dot{U}_{\ell,a} &= r_a \delta_{\ell,a} e^\top - \frac{\rho}{\tau} U_{\ell,a} A^\top. \end{aligned} \quad (12)$$

Initially, the zeroth column of $H_{\ell,a}$ is $h_{\ell-1,a}(0)$ and all other columns of $H_{\ell,a}$ and $U_{\ell,a}$ vanish.

Orthogonal projection on $[0, \tau]$ has diagonal Gram matrix $G_A = \tau \text{diag}((2k+1)^{-1})$. The reconstructed physical matrix is

$$\widehat{W}_\ell = W_{\ell,0} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{P-1} (2k+1) U_{\ell,a, :, k} H_{\ell,a, :, k}^\top. \quad (13)$$

All forward and backward responses in (12) are evaluated using the current reconstructed matrices. The first layer and readout follow their canonical dense equations evaluated at the same state. Equations (2)–(13) therefore form an autonomous finite ODE. They do not use the future or the dense trajectory.

For one sample and one internal link, the lowest moments make the meaning transparent:

$$H_{:,0}(t) = h(0) + \int_0^t \rho(s) h(s) \, ds, \quad qquad U_{:,0}(t) = \int_0^t r(s) \delta(s) \, ds. \quad (14)$$

Higher modes distinguish where and how the responses changed along accumulated activity. They are history projections, not Taylor derivatives at initialization.

The learned correction in (13) has rank at most mP per internal layer. The action on a vector v is evaluated without forming a dense learned matrix:

$$(\widehat{W}_\ell - W_{\ell,0})v = -\frac{2}{m\tau} \sum_{a,k} \frac{2k+1}{n} U_{\ell,a, :, k} (H_{\ell,a, :, k}^\top v). \quad (15)$$

The action of $W_{\ell,0}$ and the corresponding transpose action remain present exactly.

3.3 Variant B: response-arclength clock

The activity clock controls the residual factor but does not directly control the speed of every stored response. Concatenate all histories that enter the learned internal matrices:

$$\Psi = \text{concat}_{\ell=2}^L \text{concat}_{a=1}^m (h_{\ell-1,a}, b_{\ell,a}). \quad (16)$$

On an interval with $\rho > 0$, define

$$\dot{\tau} = g := \rho + \left\| \dot{\Psi} \right\|_2, \quad qquad \tau(0) = 1. \quad (17)$$

This is a causal function of the current closure state. Its key property is

$$\left\| \frac{d\Psi}{d\tau} \right\|_2 = \frac{\left\| \dot{\Psi} \right\|_2}{\rho + \left\| \dot{\Psi} \right\|_2} \leq 1. \quad (18)$$

The clock controls the joint path speed; it does not equalize every neuron and is not claimed to minimize finite-order projection error.

Changing coordinates must not change the learning integral. We therefore advance the location of history at speed g but insert mass at speed ρ . For a test function v define

$$\int v(\xi) \, d\mu_t(\xi) = \int_0^1 v(\xi) \, d\xi + \int_0^t v(\tau(s)) \rho(s) \, ds. \quad (19)$$

On the physical-history portion, $d\mu_t/d\xi = \rho/g \leq 1$. Both prefix histories now match their initial values: $h(\xi) = h(0)$ and $b(\xi) = b(0)$ for $0 \leq \xi \leq 1$. The known prefix product is subtracted from the reconstruction so it causes no artificial learning.

Define weighted moments and one shared $P \times P$ Gram matrix,

$$\begin{aligned} H_{\ell,a} &= \int h_{\ell-1,a}(\xi) p(\xi/\tau)^\top d\mu_t(\xi), \\ U_{\ell,a} &= \int b_{\ell,a}(\xi) p(\xi/\tau)^\top d\mu_t(\xi), \\ G &= \int p(\xi/\tau) p(\xi/\tau)^\top d\mu_t(\xi). \end{aligned} \quad (20)$$

They satisfy

$$\begin{aligned} \dot{H}_{\ell,a} &= \rho h_{\ell-1,a} e^\top - \frac{g}{\tau} H_{\ell,a} A^\top, \\ \dot{U}_{\ell,a} &= r_a \delta_{\ell,a} e^\top - \frac{g}{\tau} U_{\ell,a} A^\top, \\ \dot{G} &= \rho e e^\top - \frac{g}{\tau} (AG + GA^\top). \end{aligned} \quad (21)$$

At initialization, only the zeroth columns of H and U are nonzero and $G = \text{diag}((2k+1)^{-1})$. The reconstruction is

$$\widehat{W}_\ell = W_{\ell,0} - \frac{2}{nm} \sum_{a=1}^m \left[U_{\ell,a} G^{-1} H_{\ell,a}^\top - b_{\ell,a}(0) h_{\ell-1,a}(0)^\top \right]. \quad (22)$$

The prefix makes G positive definite at every fixed finite order and finite clock length. In computation one solves linear systems with G rather than forming G^{-1} . The correction rank is at most $m(P+1)$.

3.4 Why the new clock is explicit

At first sight (17) may appear implicit because $\dot{\Psi}$ depends on matrix velocities. Differentiating (22) resolves the issue. Set

$$h_{\ell-1,a}^* = H_{\ell,a} G^{-1} e, \quad b_{\ell,a}^* = U_{\ell,a} G^{-1} e. \quad (23)$$

Every dilation term proportional to g/τ cancels, leaving

$$V_\ell := \dot{\widehat{W}}_\ell = -\frac{2\rho}{nm} \sum_a \left[b_{\ell,a}(h_{\ell-1,a}^*)^\top + b_{\ell,a}^* (h_{\ell-1,a} - h_{\ell-1,a}^*)^\top \right]. \quad (24)$$

This velocity is known before g . Set $V_1 = F_1(\widehat{\theta})$ and $V_w = F_w(\widehat{\theta})$ and differentiate the forward pass:

$$\begin{aligned} \dot{z}_{1,a} &= V_1 u_a, \\ \dot{z}_{\ell,a} &= V_\ell h_{\ell-1,a} + \widehat{W}_\ell \dot{h}_{\ell-1,a}, \\ \dot{h}_{\ell,a} &= \phi'_\ell(z_{\ell,a}) \odot \dot{z}_{\ell,a}, \\ \dot{r}_a &= \frac{1}{n} (V_w^\top h_{L,a} + w^\top \dot{h}_{L,a}), \quad \dot{\rho} = \frac{1}{m\rho} \sum_a r_a \dot{r}_a. \end{aligned} \quad (25)$$

Differentiate backpropagation from layer L down to layer two:

$$\begin{aligned}
\dot{\delta}_{L,a} &= V_w \odot \phi'_L(z_{L,a}) + w \odot \phi''_L(z_{L,a}) \odot \dot{z}_{L,a}, \\
\dot{\delta}_{\ell,a} &= \phi''_{\ell}(z_{\ell,a}) \odot \dot{z}_{\ell,a} \odot \widehat{W}_{\ell+1}^{\top} \delta_{\ell+1,a} \\
&\quad + \phi'_{\ell}(z_{\ell,a}) \odot \left(V_{\ell+1}^{\top} \delta_{\ell+1,a} + \widehat{W}_{\ell+1}^{\top} \dot{\delta}_{\ell+1,a} \right), \\
\dot{b}_{\ell,a} &= \frac{\dot{r}_a \delta_{\ell,a} + r_a \dot{\delta}_{\ell,a}}{\rho} - b_{\ell,a} \frac{\dot{\rho}}{\rho}.
\end{aligned} \tag{26}$$

These formulas determine $\dot{\Psi}$, then g , and finally (21). The algorithm uses directional derivatives, fixed matrix actions, moment contractions, and a small Gram solve. It uses no full Jacobian, Hessian, future trajectory, or dense learned matrix. Notice that ϕ''_1 is unnecessary: the differentiated backward histories begin at layer two.

If $\rho(0) = 0$, both dense and closure trajectories are stationary; we define the closure to remain there without evaluating r/ρ . The theorem below also shows that a nonstationary regular trajectory cannot reach zero residual at finite time under its hypotheses.

4 Finite-horizon approximation theorem

Throughout this section, n, d, m, L , the data, and the initialization are fixed. The norm of a physical state is the Euclidean product of Frobenius norms for all matrices and the Euclidean norm for the readout. Put $\lambda_P = P(P+1)$.

4.1 Polynomial approximation and the exact defect

Lemma 3 (Shifted-Legendre projection). *Let $v : [0, \tau] \rightarrow \mathbb{R}^q$ be absolutely continuous with $v' \in L^2$. If $\Pi_P v$ is its ordinary L^2 projection onto polynomials of degree less than P , then*

$$\|v - \Pi_P v\|_{L^2(0,\tau)}^2 \leq \frac{\tau^2}{4\lambda_P} \|v'\|_{L^2(0,\tau)}^2. \tag{27}$$

If μ is any measure with $d\mu \leq d\xi$, the best degree- $(P-1)$ projection in $L^2(\mu)$ has error no larger than the right-hand side of (27). A degree- $(P-1)$ polynomial q also obeys

$$\|q(\tau)\|_2 \leq \frac{P}{\sqrt{\tau}} \|q\|_{L^2(0,\tau)}. \tag{28}$$

Proof. The shifted Legendre equation is

$$-\frac{d}{dx} [x(1-x)p'_k(x)] = k(k+1)p_k(x). \tag{29}$$

Integration by parts and (7) give

$$\int_0^1 x(1-x)p'_k p'_j dx = \frac{k(k+1)}{2k+1} \mathbf{1}_{j=k}. \tag{30}$$

For the Legendre coefficients $c_k = (2k+1) \int_0^1 v(x)p_k(x) dx$, Bessel's inequality therefore yields

$$\sum_{k \geq 1} \frac{k(k+1)}{2k+1} \|c_k\|_2^2 \leq \int_0^1 x(1-x) \|v'(x)\|_2^2 dx. \tag{31}$$

Parseval's identity, $k(k+1) \geq \lambda_P$ in the omitted tail, and $x(1-x) \leq 1/4$ prove the unit-interval result. Rescaling $x = \xi/\tau$ proves (27). For a smaller measure, the ordinary projection is an admissible comparison polynomial and weighted best approximation can only improve its weighted error. Finally expand $q = \sum_{k < P} c_k p_k(\xi/\tau)$, use $p_k(1) = 1$, and apply Cauchy–Schwarz with $\sum_{k < P} (2k+1) = P^2$ to obtain (28). $\square$

For either closure let Π_t be its current projection in the corresponding history measure. For an inserted vector history f , define

$$D_f(t) = \|f - \Pi_t f\|_{L^2(\mu_t)}^2, \text{quad} f^* = (\Pi_t f)(\tau(t)). \quad (32)$$

Lemma 4 (Exact accumulator and projection-energy identities). *Along the closure's own response history,*

$$\dot{D}_f = \rho \|f_{\text{current}} - f^*\|_2^2, \text{quad} D_f(0) = 0. \quad (33)$$

The closure's physical state satisfies

$$\dot{\hat{\theta}} = F(\hat{\theta}) + E, \text{quad} E_1 = E_w = 0, \quad (34)$$

where, for $2 \leq \ell \leq L$,

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_{\ell,a} - b_{\ell,a}^*) (h_{\ell-1,a} - h_{\ell-1,a}^*)^\top. \quad (35)$$

Consequently,

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell,a}(t) D_{h,\ell,a}(t)}. \quad (36)$$

Equivalently, if

$$W_\ell^{\text{acc}}(t) = W_{\ell,0} - \frac{2}{nm} \sum_a \int_0^t \hat{r}_a \hat{\delta}_{\ell,a} \hat{h}_{\ell-1,a}^\top ds, \quad (37)$$

then $R_\ell = \hat{W}_\ell - W_\ell^{\text{acc}}$ is a proof-only accumulator discrepancy with $\dot{R}_\ell = E_\ell$.

Proof. Writing the projection through its moments gives $\|\Pi_t f\|^2 = \text{tr}(B_f G^{-1} B_f^\top)$. Differentiate this expression using the moment and Gram equations. The dilation terms cancel, and the insertion terms give

$$\frac{d}{dt} \|\Pi_t f\|^2 = \rho \left[2 \langle f_{\text{current}}, f^* \rangle - \|f^*\|^2 \right]. \quad (38)$$

The derivative of the raw history energy is $\rho \|f_{\text{current}}\|^2$, proving (33). Differentiating the matrix reconstructions and subtracting the dense vector field at the closure state gives (35). The Frobenius norm of $uv^\top$ is $\|u\| \|v\|$; Cauchy–Schwarz in activity measure and (33) give (36).

The same cancellation has a static interpretation. Orthogonality gives

$$\int b h^\top d\mu - \int (\Pi_t b)(\Pi_t h)^\top d\mu = \int (b - \Pi_t b)(h - \Pi_t h)^\top d\mu. \quad (39)$$

Both mixed terms vanish. The error in the learned interaction is therefore a product of the two projection residuals. $\square$

4.2 Main theorem

Theorem 5 (Global finite-horizon response-memory tracking). *Let the dense network be defined by (2)–(5), with arbitrary finite data and initial weights. For every finite $T \geq 0$:*

1. *If every $\phi_\ell \in C_{\text{loc}}^{1,1}(\mathbb{R})$, there are finite $P_A(T)$ and $C_A(T)$, independent of P , such that Variant A has a unique solution on $[0, T]$ for every $P \geq P_A(T)$ and*

$$\sup_{t \leq T} \|\hat{\theta}_P(t) - \theta(t)\| \leq \frac{C_A(T)}{\sqrt{P(P+1)}}. \quad (40)$$

2. *If $\phi_1 \in C_{\text{loc}}^{1,1}(\mathbb{R})$ and $\phi_\ell \in C_{\text{loc}}^{2,1}(\mathbb{R})$ for $2 \leq \ell \leq L$, there are finite $P_B(T), C_B(T), \Lambda_T$, independent of P , such that Variant B has a unique regular solution on $[0, T]$ for every $P \geq P_B(T)$,*

$$\sup_{t \leq T} \tau_P(t) \leq \Lambda_T, \quad \sup_{t \leq T} \|\hat{\theta}_P(t) - \theta(t)\| \leq \frac{C_B(T)}{P(P+1)}. \quad (41)$$

If the initial residual is zero, both statements use the stationary extension and hold exactly. If all activations and their first derivatives are globally bounded, Variant A exists for every physical time at every $P \geq 1$, and (40) holds at every order with $P_A(T) = 1$.

The activation class in part 1 includes tanh, sigmoid, softplus, exact GELU, SiLU, arctangent, and smooth ReLU approximations. Part 2 requires one more derivative in the upper hidden layers because it differentiates the backward responses. Exact ReLU is outside both stated smoothness theorems, even though it is included in some numerical tests.

Proof. We give the complete argument in four steps. Explicit finite constants and the all-order bounded-activation continuation are collected in Appendix A.2.

1. Dense bounds and a stopped common region. The gradient field is locally Lipschitz under the stated assumptions. If $\mathcal{M}$ is the block mobility, then

$$\frac{d}{dt} \mathcal{L} = - \left\| \mathcal{M}^{-1/2} \dot{\theta} \right\|^2, \quad \text{quad} \int_0^T \left\| \dot{\theta} \right\|^2 dt \leq n \rho(0)^2. \quad (42)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nT} \rho(0)$. A finite-time maximal endpoint would have a finite parameter limit and could be continued, so dense flow exists globally.

Choose a closed physical-state ball $\mathcal{B}$ whose boundary is distance one from the dense path on $[0, T]$, and stop a closure at its first exit. On $\mathcal{B}$ all responses are bounded. There are finite constants q, V, K, c_f such that

$$\rho \leq q, \quad \text{quad} \|F(\vartheta)\| \leq V \rho(\vartheta), \quad \text{quad} \|F(\vartheta) - F(\vartheta')\| \leq K \|\vartheta - \vartheta'\|, \quad (43)$$

and residual RMS is c_f -Lipschitz in physical state. Put $S = Tq$ and $\ell_T = 1 + S$. Along the dense path, whenever $\rho > 0$, $|\dot{\rho}| \leq c_f V \rho$, hence

$$\rho(t) \geq \mu := \rho(0) e^{-c_f V T} > 0. \quad (44)$$

2. Variant A. Use activity coordinate $\xi = \tau(t)$ on a regular stopped segment and define the mean squared derivative energies

$$Z_j(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \|\partial_\xi h_{j,a}(\xi)\|^2 d\xi. \quad (45)$$

The prefix is constant. If B_ℓ bounds $\delta_{\ell,a}$ on $\mathcal{B}$, the normalization $m^{-1} \sum_a (r_a/\rho)^2 = 1$ gives

$$\frac{1}{m} \sum_a D_{h,\ell,a} \leq \frac{\ell_T^2}{4\lambda_P} Z_{\ell-1}, \quad (46)$$

$$\frac{1}{m} \sum_a D_{b,\ell,a} \leq S B_\ell^2. \quad (47)$$

Only the forward histories need derivative control.

It remains to show that Z_j itself does not grow with P . Let $e_\ell = E_\ell/\rho$. Projection contraction, the endpoint estimate (28), and (46) yield

$$\int_0^t \rho \|e_\ell\|^2 ds \leq C_\ell Z_{\ell-1}(t), \quad (48)$$

where C_ℓ is finite and independent of P : the endpoint factor is $(P+1)^2$, while (46) contributes $1/[P(P+1)]$. This cancellation is the key old-clock estimate.

For the first layer, $\partial_\xi h_{1,a}$ is uniformly bounded because $\dot{W}_1$ contains the factor ρ . For later layers the chain rule gives

$$\partial_\xi h_{\ell,a} = \phi'_\ell(z_{\ell,a}) \odot \left[(F_\ell/\rho + e_\ell) h_{\ell-1,a} + \widehat{W}_\ell \partial_\xi h_{\ell-1,a} \right]. \quad (49)$$

Squaring, averaging, integrating in ξ , and applying (48) gives a finite recursion $Z_1 \leq \overline{Z}_1$ and $Z_\ell \leq c_\ell(1 + \overline{Z}_{\ell-1}) =: \overline{Z}_\ell$. Fixed depth therefore gives P -independent bounds for every Z_j .

Combining (36), (46), and (47) gives a finite Γ_A , independent of P , such that

$$\int_0^t \|E(s)\| ds \leq \frac{\Gamma_A}{\sqrt{\lambda_P}}. \quad (50)$$

3. Variant B. Stop additionally if closure residual reaches $\mu/2$. On this region the current-response map $\theta \mapsto \Psi(\theta)$ has bounded derivative, $\|D\Psi\|_{\text{op}} \leq J$. This follows from compactness and the formula

$$D(r/\rho)[v] = \left(I - \frac{q_r q_r^\top}{m} \right) \frac{Dr[v]}{\rho}, \quad qqquad q_r = r/\rho, \quad quad \|q_r\|^2 = m, \quad (51)$$

where the parenthesized matrix is an orthogonal projection.

First use a coarse estimate independent of order. Projection errors are no larger than raw history energies, which are bounded on $\mathcal{B}$ and have total mass at most ℓ_T . Equations (36) and $2\sqrt{uv} \leq u+v$ give

$$\int_0^t \|E\| ds \leq C_0, \quad qqquad \int_0^t \|\hat{\theta}\| ds \leq VS + C_0. \quad (52)$$

Therefore the clock has the order-independent bound

$$\tau(t) = 1 + \int_0^t \left(\rho + \|\dot{\Psi}\| \right) ds \leq \ell_T + J(VS + C_0) =: \Lambda_T. \quad (53)$$

This bound is derived; it is not an imposed clock cap.

The concatenated prefixed history is continuous, absolutely continuous, and by (18)

$$\int_0^\tau \|\partial_\xi \bar{\Psi}\|^2 d\xi \leq \tau - 1. \quad (54)$$

Because $d\mu \leq d\xi$, Lemma 3 gives

$$\sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{\tau^2(\tau - 1)}{4\lambda_P}. \quad (55)$$

Using (36) and $2\sqrt{uv} \leq u + v$ once more,

$$\int_0^t \|E(s)\| ds \leq \frac{\Gamma_B}{\lambda_P}, \quad \text{quad} \Gamma_B = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (56)$$

This is where the quadratic rate enters: both histories are first-order accurate and their paired error multiplies.

4. Feedback and continuation. Subtract dense flow from (34). On the stopped interval, with $d(t) = \|\hat{\theta}(t) - \theta(t)\|$,

$$d(t) \leq \varepsilon_P + K \int_0^t d(s) ds, \quad (57)$$

where $\varepsilon_P = \Gamma_A/\sqrt{\lambda_P}$ for Variant A and $\varepsilon_P = \Gamma_B/\lambda_P$ for Variant B. Grönwall's inequality gives

$$d(t) \leq \varepsilon_P e^{KT}. \quad (58)$$

Choose P so this is less than $1/2$. Then the closure cannot reach the physical boundary, which is distance one from the dense path. For Variant B, also require $c_f \varepsilon_P e^{KT} \leq \mu/4$; then its residual remains above $3\mu/4$ and cannot reach the stopping boundary $\mu/2$.

The prefix makes G positive definite. At fixed P , its least eigenvalue has a positive minimum for $1 \leq \tau \leq \Lambda_T$. Moment integral representations bound every internal coordinate on the stopped interval. Thus the full finite state stays in a compact subset of its locally Lipschitz ODE domain and can be continued to T . This contradicts any earlier stopped endpoint and proves (40)–(41).

When activations and first derivatives are globally bounded, a downward induction from the readout bounds every internal reconstructed matrix for Variant A independently of P and time on each finite interval. The same moment continuation then works at every order. Appendix A.2 gives the induction explicitly. $\square$

Corollary 6 (Outputs, losses, and order selection). *On any bounded passive-input set $\mathcal{X}$, the same common physical-state region has a finite output Lipschitz constant c_{test} . Hence*

$$\sup_{x \in \mathcal{X}, t \leq T} |\hat{f}_P(t, x) - f(t, x)| \leq c_{\text{test}} \sup_{t \leq T} \|\hat{\theta}_P(t) - \theta(t)\|. \quad (59)$$

Training residual and loss discrepancies obey the same order rates. A sufficient order for physical-state tolerance ε is

$$\begin{aligned} \text{Variant A: } P &\geq P_A(T), & P(P+1) &\geq [C_A(T)/\varepsilon]^2, \\ \text{Variant B: } P &\geq P_B(T), & P(P+1) &\geq C_B(T)/\varepsilon. \end{aligned} \quad (60)$$

The constants are certificates, not predictions of an economical practical order. They can grow rapidly with T , depth, width, and response bounds. The theorem does not establish $\sup_{t \geq 0} \|\hat{\theta}_P(t) - \theta(t)\| \rightarrow 0$.

Table 1: Leading storage and algebraic cost for L hidden layers of width n and a full batch of m samples. Constants and first/readout-layer terms are omitted. “Fixed action” is the multiplication by the retained $W_{\ell,0}$ and its transpose.

Quantity	Dense flow	Activity clock	Response clock
Moving internal state	$(L-1)n^2$	$2(L-1)nmP$	$2(L-1)nmP + P^2$
Fixed initialized state	none beyond current weights	$(L-1)n^2$	$(L-1)n^2$
Learned-correction rank	at most n	at most mP	at most $m(P+1)$
Fixed forward/backward actions, full batch	included in dense actions	$O((L-1)n^2m)$	$O((L-1)n^2m)$
Learned factor actions, full batch	$O((L-1)n^2m)$ total dense work	$O((L-1)nm^2P)$	$O((L-1)nm^2P)$ plus solves
Small-order algebra	—	$O((L-1)nmP)$ with cumulative triangular sums	$O(P^3)$ Gram factorization and up to $O((L-1)nmP^2)$ factor solves

4.3 What follows at the population limit

Corollary 7 (Diagonal population consistency). *Suppose a sequence of dense width- n networks satisfying the model above has observables $\mathcal{O}_n$ that converge uniformly on $[0, T]$ in probability to a population-flow observable $\mathcal{O}_\infty$. Then there exists a width-dependent sequence $P(n) \rightarrow \infty$ such that the corresponding response-memory observables converge to $\mathcal{O}_\infty$ in probability on $[0, T]$.*

Proof. For each fixed n , Theorem 5 and the observable Lipschitz bound allow an order $P(n)$ large enough that the closure–dense discrepancy is at most $1/n$. The triangle inequality with dense–population convergence proves the claim. $\square$

This sequential bridge is the population statement presently justified. It does not bound $P(n)$, prove convergence at fixed P as $n \rightarrow \infty$, or give a width-uniform error constant. Establishing those properties is a separate problem because the closure retains the reused initialized operators and the response-clock norm itself scales with the size of the concatenated particle state.

5 State, rank, and computational cost

The closures remove growth with elapsed training time. They do not remove every dependence on width or sample count. Table 1 separates moving learned state from retained initialized state.

For Variant A, the triangular dilation sums can be evaluated by cumulative prefix sums in $O(P)$ per neuron history. For Variant B, a Cholesky factorization of the shared Gram matrix costs $O(P^3)$; solving for all factor coordinates can cost $O((L-1)nmP^2)$ in a direct implementation. The response monitor adds directional forward and backward passes. Conditioning of G can deteriorate with P even though the prefix proves it remains positive definite at every fixed order.

The dominant fixed-matrix actions remain quadratic in width. Consequently, the present method does not give a subquadratic end-to-end evaluation of a generic dense initialization. It can reduce the cost of storing and applying the *learned correction* when $mP \ll n$, and it can avoid materializing a dense updated matrix. A dense implementation that stores only its current weights may still use less minimal total memory because the closure stores both W_0 and its moving history state.

The theorem turns order into a conservative state–accuracy tradeoff. Ignoring constants and thresholds, Variant A requires $P = O(\varepsilon^{-1})$ and hence $O((L-1)nm\varepsilon^{-1})$ moving history coordinates.

Table 2: Activity-clock RMS discrepancy from the dense fitted circle function. Fresh dense references are used where available. The two-outlier row retains the study’s selected archived reference. The smallest P7 values are near or below measured numerical-refinement sensitivity and should be read at their order of magnitude.

Task	$P = 1$	$P = 3$	$P = 7$
Two outliers, alternating	0.236313	0.0729531	0.0046631
Quadrant, alternating	0.341318	0.0456113	0.0015523
Quadrant, paired labels	0.0192813	0.0028176	0.0001185
Quadrant, center/edges	0.0455816	0.0048778	0.0013076
Equal mixed odd	0.0123995	0.0005257	3.7×10^{-6}

Variant B requires $P = O(\varepsilon^{-1/2})$, giving $O((L - 1)nm\varepsilon^{-1/2})$ history coordinates plus a $P^2 = O(\varepsilon^{-1})$ shared Gram matrix. These are upper-bound scalings, not observed asymptotic rates. At small practical orders the extra clock and Gram conditioning can make Variant B less accurate or slower than Variant A.

At large width, each moment column is naturally represented by one value per neuron. This is the particle implementation of a population field. The current canonical implementation evaluates the learned action through neuron population averages such as

$$\frac{1}{n} H_{\ell,a,:,k}^\top h_{\ell-1,b}, \quad (61)$$

and uses the same factors for the transpose action. Forward and adjoint reuse of one initialized operator is preserved exactly; redrawing or replacing it by independent samples would change the training law.

The empirical memory measurements reinforce the distinction. In the width-4096 MNIST100 experiment, history factors occupy 6.25, 12.50, and 18.75 MiB for $P = 1, 2, 3$, compared with 128 MiB for an unrestricted learned middle matrix. The retained W_0 adds another 128 MiB, so minimal moving-plus-fixed state is slightly larger than the dense current-state baseline. The tested closures nevertheless had lower measured CUDA peaks because their solver workspaces avoided dense evolving matrix stages; they were slower at the common tolerances. These are implementation measurements, not an asymptotic speedup.

6 Finite experiments

The experiments ask whether small-order closures reproduce the *function learned by dense gradient flow*. They do not test only training labels. Each closure uses the same initialized weights as its dense reference and retains the same W_0 . Unless stated otherwise, models are compared at their own first crossing of the same training-loss target. This is a matched-loss endpoint comparison, not an equal-time comparison or a proof of continuous time accuracy.

6.1 Sparse-label functions on the circle

The first panel uses bias-free two-hidden-layer tanh networks of width 2048, canonical mobilities $(n, 1, n)$, eight labeled unit-circle inputs, and one initialization. The equal-mixed task has an exact four-sample antipodal quotient. Dense and activity-clock trajectories stop at their own first training-MSE 10^{-3} crossing. Predictor discrepancy is RMS over 8192 equally spaced circle directions.

Own-endpoint functions on the circle
Width 2048; each model at its own training-MSE 0.001 endpoint; closure retains dense W0

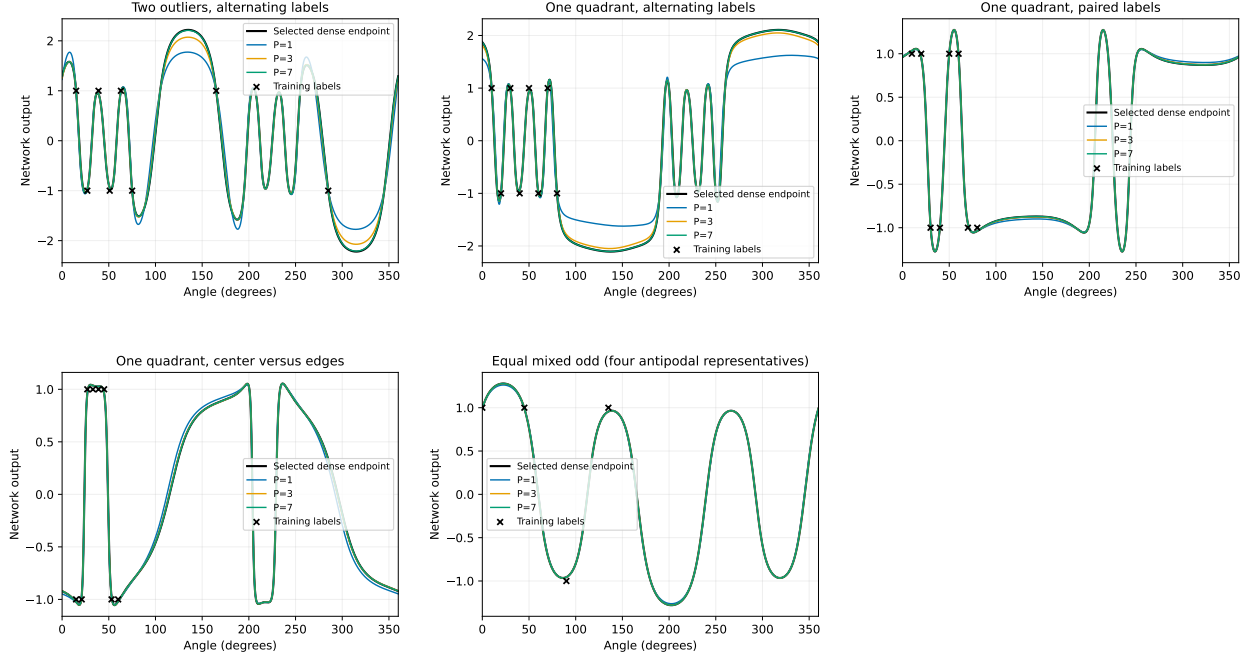


Figure 1: Dense and activity-clock fitted functions for five circle tasks. Every model is shown at its own first training-MSE 10^{-3} crossing. The closure retains the dense initialized middle matrix. The hard-outlier dense reference is inherited from the frozen benchmark; the other rows have fresh references checked within the response-memory study.

Figure 1 shows the full learned functions. The training points occupy only a small part of the evaluation grid, so agreement between curves is stronger evidence than common training fit. There is no ground-truth label function on the unlabeled circle; these are approximation errors against dense learning, not generalization errors.

Figure 2 gives the requested radial view for a fully in-study fresh-reference task. Network output is signed, so the plotted radius is explicitly shifted by two; radial tick labels report the original output.

The same construction was tested with three hidden tanh layers of width 4096, so two dense internal matrices were compressed simultaneously. Every run reached MSE 10^{-3} and passed the declared numerical gates. Results in Table 3 show strong improvement over $P = 1$, but also two resolved $P = 2$ versus $P = 3$ reversals.

6.2 Evolving response memory versus fixed or directly trained factors

The response moments are not a frozen dictionary. Their columns change with the current neural responses, and the reconstructed network feeds back into their derivatives. In the historical fixed-dictionary experiments, learned increments were confined to spans chosen at initialization. Such methods could fit coefficients but could not acquire new response directions. On the same difficult circle functions they were much less efficient in the reported vector counts. Those counts do not match total storage because response memory retains W_0 .

A cleaner control inside the current study trains a direct factorization $W = W_0 + AB$ by its

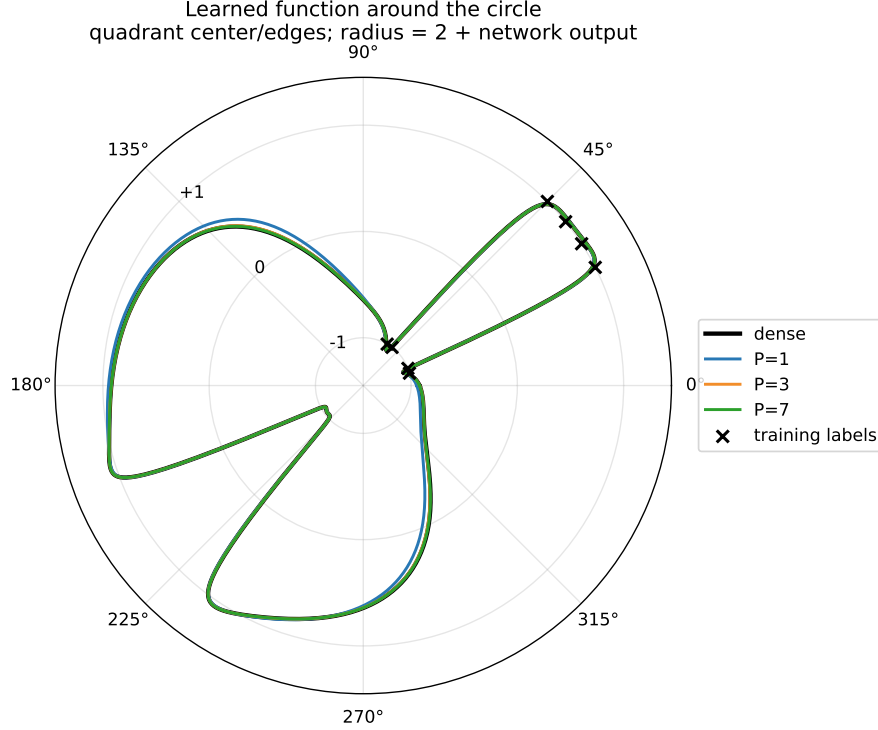


Figure 2: Polar rendering of the quadrant center/edges experiment. The angle is the input direction and radius is $2 + f(x)$. Black crosses are training labels. The figure is postprocessed from frozen predictions and involves no new training.

own Euclidean factor gradient flow, with matched correction rank and two factor initializations. Response memory has lower function RMS in all 29 numerically valid paired cells; 28 differences exceed a factor of two. At rank 24 on the alternating-outlier task, for example, $P = 3$ response memory has RMS 0.07295, versus 0.46546 and 0.46091 for the two direct-factor runs. At rank 56, the response error is 0.00466 versus 0.51621 and 0.25103. This rejects the explanation that any low-rank learned correction would perform equally well under its natural factor flow. It does not prove optimality over all low-rank algorithms.

6.3 MNIST predictions outside the training set

The MNIST experiment uses digits 3 and 8, 50 training images per class, and all 1984 official-test images of those digits for evaluation. Inputs are normalized per image to unit Euclidean norm; no test image enters training, stopping, or numerical control. The network has two tanh hidden layers of width 4096. At each model’s first training-MSE 10^{-3} crossing, held-out RMS discrepancy from dense predictions is

$$0.00324756 \ (P = 1), \quad 0.00108443 \ (P = 2), \quad 0.00119922 \ (P = 3). \quad (62)$$

All declared refinement gates pass. The small $P = 3$ worsening relative to $P = 2$ is resolved under the frozen diagnostic rule. All models have the same held-out classification accuracy, but prediction agreement in Figure 4 is the relevant closure evidence.

The history factors occupy 6.25, 12.50, and 18.75 MiB, while an unrestricted middle correction has 128 MiB. Adding the retained 128 MiB initialized matrix makes minimal total closure state

Table 3: Three-hidden-layer circle RMS discrepancy from dense learning at matched MSE 10^{-3} .

Task	$P = 1$	$P = 2$	$P = 3$
Two outliers, alternating	0.108086	0.023797	0.031476
Quadrant, alternating	0.274643	0.028358	0.058491
Quadrant, paired labels	0.012186	0.003708	0.001215
Quadrant, center/edges	0.078973	0.025702	0.007246
Equal mixed odd	0.014985	0.002412	0.000282

Dense-function RMS at each model's own training-MSE 0.001 endpoint
Width 2048; identical fixed W0; prescribed factor seeds reported separately

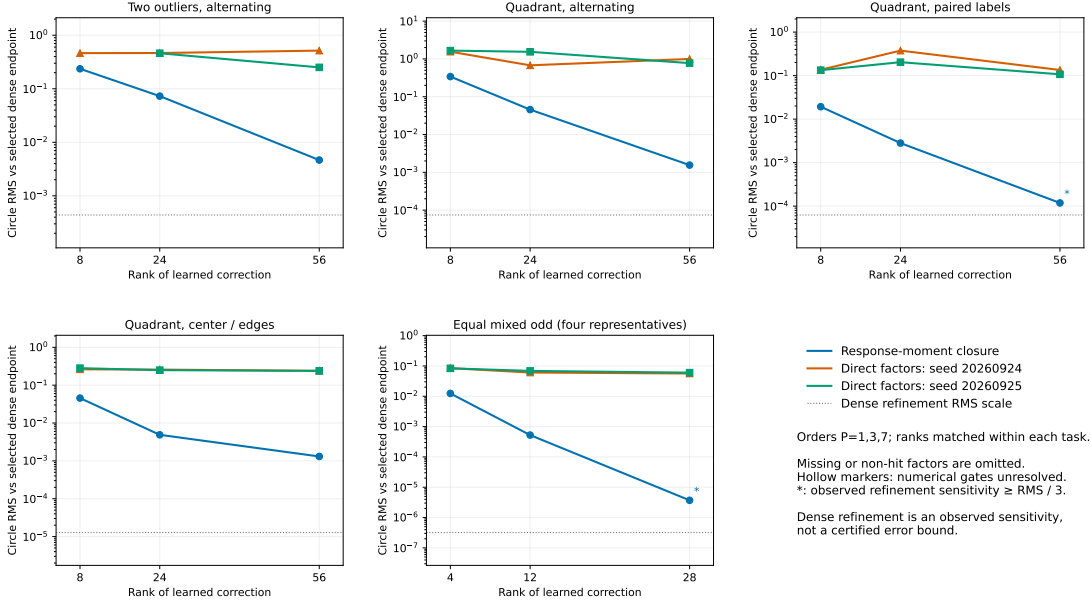


Figure 3: Response-memory closure versus direct factor-gradient-flow controls at matched structural rank. Each model is compared with the same dense fitted function. Two factor seeds are shown; one capped factor cell is excluded from fitted-endpoint comparison.

slightly larger than the dense current state. Finest-tolerance integration took 103.0 seconds for dense flow and 139.9, 149.9, and 152.3 seconds for the closures. Measured CUDA peaks were lower for the closures, but the experiment does not establish a hardware-independent speedup.

6.4 The sharper clock: promise and adverse cases

The response clock has the stronger asymptotic theorem but need not dominate at low order. In a focused width-2048 fixed-step panel, all old- and new-clock models reached training RMS 0.05. On the tanh alternating-quadrant task, the new clock improves the $P = 5$ full-circle discrepancy from 0.00658 to 0.00346. On GELU alternating outliers it improves old-clock $P = 5$ 0.00196 to 0.00098. On GELU alternating quadrant, however, its $P = 5$ error is 0.99568, versus 0.01279 for the activity clock. Halving the maximum step reduces the new-clock error to 0.10313, revealing substantial finite-step sensitivity without eliminating the gap.

A broader single-seed response-clock sweep contains 82 activation/depth/task configurations and 328 fits, each capped at 20 seconds. Dense flow fits 79 configurations; $P = 1, 2, 3$ fit 70, 67, and 66.

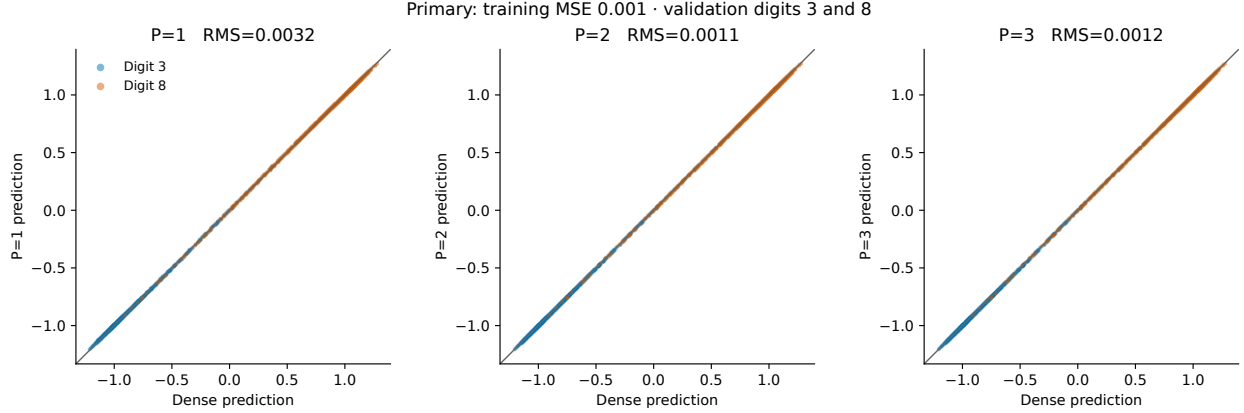


Figure 4: Held-out closure predictions against dense predictions on 1984 MNIST 3/8 test images. Models stop at their own training-MSE 10^{-3} endpoint. The correction-rank bounds are 100, 200, and 300 for $P = 1, 2, 3$.

On the same 65 configurations where all four models fit, median closure–dense test RMS is 0.03147, 0.01339, and 0.00658. The largest fitted $P = 3$ discrepancies remain 1.36, 1.25, and 0.37 on three alternating-quadrant cases. The panel supports typical improvement with order and documents clear exceptions. It does not empirically verify an $O(P^{-2})$ rate.

All headline experiments use one initialization, and MNIST uses one selected training subset. Numerical refinement differences are sensitivity checks, not rigorous error bars. Appendix B records the full old/new-clock table and resource accounting used here.

7 Relation to prior work and remaining limits

7.1 Exact wide-network descriptions carry history

Tensor Programs rigorously identify discrete-time infinite-width feature-learning limits at each fixed finite training horizon [Yang and Hu, 2021]. In a generic deep nonlinear network, the exact construction introduces Gaussian variables, covariances, and transpose response terms indexed by previous steps. Yang and Hu describe a generic quadratic-in-step storage and computation burden for this representation. Recent work proves rich feature nondegeneracy throughout deep μP training, while its recursive state continues to contain historical dependencies [Chen et al., 2025]. These results establish that the standard exact representation grows with its horizon; they do not prove that every exact realization must have growing memory.

Self-consistent neural-network DMFT organizes infinite-width feature learning through two-time activation, gradient, and causal response kernels [Bordelon and Pehlevan, 2022]. There are also rigorous DMFT convergence theorems for random-design high-dimensional first-order methods [Celentano et al., 2026, Gerbelot et al., 2024]. Those theorems take sample and ambient dimensions to infinity proportionally while keeping a finite internal learner dimension; they do not directly identify the fixed-data, all-hidden-width deep regime here. It would therefore be wrong to describe DMFT as uniformly nonrigorous, just as it would be wrong to import a random-design convergence theorem into this network without checking its regime.

Rigorous multilayer mean-field limits and finite-width couplings are known under mean-field parametrizations and paper-specific initialization structures [Nguyen and Pham, 2023, Araújo et al., 2019]. Earlier measure-gradient-flow work gives foundational shallow particle limits [Chizat and

Bach, 2018, Sirignano and Spiliopoulos, 2020]. Our theorem addresses a different question: it compresses the learned dense increment along a fixed realized nonlinear gradient-flow trajectory, then controls the induced feedback error.

7.2 Polynomial memory and low-rank training

HiPPO is direct prior art for the central memory mechanism [Gu et al., 2020]. It derives online ODEs for optimal polynomial projections of a signal history, including expanding-interval Legendre coordinates and approximation bounds. We do not claim to introduce online Legendre memory. The additional ingredients here are the identification of paired forward/backward neural histories, their reconstruction of every internal learned matrix, an autonomous self-consistent training loop, and a finite-horizon stability proof for the perturbed trajectory.

Retaining W_0 and representing $W_t - W_0$ in low rank also has precedent. InRank studies cumulative low-rank updates and proves modewise behavior in a restricted deep-linear setting [Zhao et al., 2024]; invariant-subspace compressions have rigorous deep-linear results [Kwon et al., 2024, Yaras et al., 2024]. The response factors here are neither assumed to lie in a fixed subspace nor optimized as free low-rank parameters. They are causal summaries of the dense update’s actual response sources, and their directions evolve as the network learns.

Among the primary sources reviewed for this draft, we did not find a theorem with the following conjunction: a fixed-order Legendre closure of coupled forward/backward response histories for deep nonlinear neural-network gradient flow, together with a finite-horizon trajectory error bound for the closed system at fixed finite width. This is a search-qualified distinction, not an unconditional priority claim.

7.3 What the result does and does not compress

The strongest exact statement is deliberately narrow:

- P is fixed throughout one closure run, so state does not grow with elapsed time.
- The learned dense correction is replaced by population moment fields and has a finite-rank action.
- The initialized hidden matrices are retained as fixed operators, in both forward and transpose orientations.
- Approximation is global on every prescribed finite horizon after choosing sufficiently large P ; it is not uniform over all $t \geq 0$.
- The theorem is fixed-width. Corollary 7 gives a width-dependent diagonal bridge, not a fixed-order population theorem.

These distinctions explain why the result can be both a real training compression and an incomplete model compression. It removes the dense *moving* internal matrices and unbounded historical state. It does not remove the random operator that repeatedly mixes neuron populations. A future width-uniform theory must either characterize that reused operator by a causal population law or prove an effective-rank approximation compatible with both orientations.

The sample index is another unresolved axis. Current moving state grows as mP per layer. Arbitrary labels preclude a universally sample-independent exact state even in generic fixed-kernel dynamics. Structured input laws may permit a second approximation, such as Fourier–Galerkin compression of the sample-indexed population fields, but that is separate from the theorem here.

7.4 Open mathematical and empirical questions

Several questions remain concrete.

1. **Width-uniform estimates.** The constants in Theorem 5 may depend strongly on n . The Euclidean response-arclength clock also scales with the concatenated neuron state. A population-normalized clock and width-uniform stability estimate are needed for a direct $n \rightarrow \infty$ theorem at controlled order.
2. **All-time stability.** The current bounds contain finite-horizon clock lengths and a Grönwall factor. A uniform all-time result would need finite total activity or a trained propagator whose integrated amplification remains bounded.
3. **Initialized-operator compression.** Compressing W_0 while preserving its correlated reuse and its adjoint is essential for a true total-memory improvement. Fresh random redraws do not preserve the target dynamics.
4. **Nonsmooth and normalized networks.** Exact ReLU, normalization layers, biases, convolution, residual connections, stochastic gradients, and discrete optimizers require new analysis. Existing numerical tests outside the theorem show both successes and failures.
5. **Practical clock design.** The response clock proves a better asymptotic order but introduces a Gram system and can behave worse at small P . A clock optimized for projection error, numerical conditioning, and population scaling remains open.
6. **Statistical validation.** The present experiments mostly use one initialization and matched-loss endpoints. Seed distributions, continuous-time solver certificates, width trends, and broader tasks are needed before making general performance claims.

At a conceptual level, a deterministic population learning law could play a role analogous to an idealized model of computation: finite networks would be understood through the ideal law plus a quantitative finite-resource error. That analogy is a research program rather than a conclusion of this paper. The response-memory theorem supplies one missing component—a controlled finite description of learned history at fixed width—while leaving the operator and population-limit steps visible.

A Supplementary proofs and explicit constants

A.1 Restricted scalar nonclosure

We prove Proposition 2 in the contraction class stated there. Consider four trainable deep-linear factors (u, B, R, v) and output

$$f_n = v^\top R B u. \tag{63}$$

Width normalizations do not affect the graph argument; they can be absorbed into width-dependent scalar coefficients. A complete contraction is encoded by a typed graph whose tensor slots are paired and summed. Multiplication of contractions is disjoint union of graphs.

Lemma 8 (Stable contraction independence). *Every fixed finite family of pairwise nonisomorphic typed complete-contraction graphs gives linearly independent contraction polynomials for every sufficiently large width.*

Proof. Treat tensor entries as independent commuting indeterminates. For every graph G , partition all index assignments according to which index vertices are equal. This writes its contraction as

$$P_G^{(n)} = \sum_{\pi} I_{G/\pi}^{(n)}, \quad (64)$$

where $I_H^{(n)}$ sums only injective labelings of the quotient incidence network H . Once width exceeds the number of vertices of each type, an injective labeling produces a tensor-entry monomial that records the typed incidence network. It occurs in another injective contraction only for an isomorphic network. Hence distinct injective contractions are independent. In a purported relation among the $P_G^{(n)}$, choose a graph with a maximal number of index vertices. Its discrete-partition injective term cannot be cancelled by any proper quotient, which has fewer vertices, or by a nonisomorphic graph of the same size. Descending in vertex count proves that all coefficients vanish. $\square$

Let $A = RB$ and $H = A^\top A$. Retain only differentiation of the endpoint factors in the feature derivative:

$$\mathcal{D}_{0,n}v = Au, \text{quad}\mathcal{D}_{0,n}u = A^\top v, \text{quad}\mathcal{D}_{0,n}A = 0. \quad (65)$$

For $j \geq 0$ put

$$U_j = v^\top AH^j u, \text{quad} X_j = u^\top H^{j+1} u, \text{quad} Y_j = v^\top (AA^\top)^{j+1} v. \quad (66)$$

Direct differentiation gives

$$\mathcal{D}_{0,n}U_j = X_j + Y_j, \text{quad}\mathcal{D}_{0,n}X_j = \mathcal{D}_{0,n}Y_j = 2U_{j+1}. \quad (67)$$

Therefore, for $j \geq 1$,

$$\mathcal{D}_{0,n}^{2j-1} f_n = 4^{j-1} \left[u^\top H^j u + v^\top (AA^\top)^j v \right]. \quad (68)$$

In the full feature derivative, every tensor replacement and product-rule coefficient is nonnegative. Hence the connected contraction

$$\Gamma_j = u^\top (B^\top R^\top RB)^j u \quad (69)$$

appears with positive coefficient in $\mathcal{D}_n^{2j-1} f_n$. Its tensor degree is $4j + 2$, so the connected graph types Γ_j have unbounded complexity.

Now suppose a fixed encoder uses contractions from a finite bounded-degree alphabet $\mathcal{S}$ and evolves by a polynomial autonomous ODE with a polynomial output readout. Every time derivative of the output obtained from that ODE is a polynomial in encoder coordinates. Its contraction graph is therefore a disjoint union of connected components from $\mathcal{S}$. Choose j so that $\Gamma_j \notin \mathcal{S}$. Equation (68) and stable independence contradict the equality between the true and encoded $(2j - 1)$ st output derivatives. An identity on a nonempty open state set is a polynomial identity, so the same coefficient comparison applies globally.

For an analytic finite-jet closure near zero, scale every factor by a common λ and compare the homogeneous Taylor coefficient of the degree of Γ_j . At any fixed power of λ , only finitely many Taylor monomials contribute, and their connected components still lie in $\mathcal{S}$. The same contradiction follows.

For squared-loss physical flow, the vector field is a scalar multiple of the feature field,

$$\mathcal{X}_n = -2\kappa(f_n - y)\mathcal{D}_n. \quad (70)$$

If $y \neq 0$, the lowest-degree part of $\mathcal{X}_n^k f_n$ is $(2\kappa y)^k \mathcal{D}_n^k f_n$. If $y = 0$, positivity of graph coefficients shows that the relevant derivative contains a positive multiple of $f_n^k \mathcal{D}_n^k f_n$. In either case a connected component Γ_j outside $\mathcal{S}$ remains, completing the proof.

The bounded-degree, state-universal, and regular-readout assumptions are essential. An unrestricted field can encode the entire future output germ, and the proposition makes no claim about such oracle encodings or about one special initialized trajectory.

A.2 Explicit constants for Theorem 5

This subsection records one concrete choice of the finite constants used in the main proof. It also makes clear where width, depth, and horizon enter.

Let

$$R = \|\theta_0\| + \sqrt{nT}\rho_0 + 1, \text{quad}\mathcal{B} = \{\theta : \|\theta\| \leq R\}, \text{quad}\rho_0 = \rho(0) > 0. \quad (71)$$

Put $X = \max_a \|u_a\|$, $M_0 = X$, and recursively define

$$\begin{aligned} s_\ell &= \sup_{|z| \leq RM_{\ell-1}} |\phi'_\ell(z)|, & M_\ell &= \sqrt{n} \sup_{|z| \leq RM_{\ell-1}} |\phi_\ell(z)|, \\ B_L &= Rs_L, & B_\ell &= s_\ell RB_{\ell+1} \quad (\ell < L). \end{aligned} \quad (72)$$

These give $\|h_{\ell,a}\| \leq M_\ell$ and $\|\delta_{\ell,a}\| \leq B_\ell$ on $\mathcal{B}$. Let

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2 \right)^{1/2}, & q &= RM_L/n + Y, & S &= Tq, & \ell_T &= 1 + S, \\ v_1 &= 2B_1X, & v_\ell &= 2B_\ell M_{\ell-1}/n \quad (\ell \geq 2), & v_w &= 2M_L, \\ V &= \left(v_1^2 + \sum_{\ell=2}^L v_\ell^2 + v_w^2 \right)^{1/2}, \end{aligned} \quad (73)$$

and

$$c_f = \frac{1}{n} \left[(B_1X)^2 + \sum_{\ell=2}^L (B_\ell M_{\ell-1})^2 + M_L^2 \right]^{1/2}. \quad (74)$$

Sample Cauchy–Schwarz gives $\|F\| \leq V\rho$ and the gradient of each output is bounded by c_f . A finite Lipschitz constant K for F on the compact convex ball follows from the local Lipschitz activation bounds.

For Variant A, a concrete order-independent derivative recursion is

$$\begin{aligned} \bar{Z}_1 &= S(s_1 v_1 X)^2, \\ \bar{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 M_{\ell-1}^2 + \left(R^2 + \frac{2B_\ell^2 \ell_T^2 M_{\ell-1}^2}{n^2} \right) \bar{Z}_{\ell-1} \right]. \end{aligned} \quad (75)$$

The derivation is exactly (48)–(49), using $\|x + y + z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$. One may take

$$\Gamma_A = \frac{\ell_T}{n} \sum_{\ell=2}^L B_\ell \sqrt{S \bar{Z}_{\ell-1}}, \text{quad} C_A(T) = \Gamma_A e^{KT}. \quad (76)$$

For Variant B, define

$$Q = m \sum_{\ell=2}^L (M_{\ell-1}^2 + B_\ell^2), \text{quad} C_0 = \frac{\ell_T Q}{nm}. \quad (77)$$

On the stopped region $\rho \geq \mu/2$, let J be a bound for the derivative of the current response map $\theta \mapsto \Psi(\theta)$. Then

$$\Lambda_T = \ell_T + J(VS + C_0), \text{quad}\Gamma_B = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}, \text{quad}C_B(T) = \Gamma_B e^{KT}. \quad (78)$$

Choose $P_A(T)$ so $\Gamma_A e^{KT} / \sqrt{P(P+1)} \leq 1/2$. Choose $P_B(T)$ so both

$$\frac{\Gamma_B e^{KT}}{P(P+1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_B e^{KT}}{P(P+1)} \leq \frac{\mu}{4}. \quad (79)$$

These choices close the first-exit arguments.

A.3 All orders for bounded activations

Suppose $\|\phi_\ell\|_\infty$ and $\|\phi'_\ell\|_\infty$ are finite. Put $M_\ell = \sqrt{n} \|\phi_\ell\|_\infty$ and $s_\ell = \|\phi'_\ell\|_\infty$. The readout equation gives the exact estimate

$$\frac{d}{dt} \|w\|^2 = -\frac{4n}{m} \sum_a (f_a - y_a) f_a \leq nY^2. \quad (80)$$

Thus $B_w = (\|w_0\|^2 + nY^2T)^{1/2}$ bounds the readout. Let

$$q = B_w M_L / n + Y, \text{quad}S = Tq, \text{quad}\ell_T = 1 + S, \text{quad}B_L = B_w s_L. \quad (81)$$

Proceed downward from layer L to layer two:

$$D_\ell = \|W_{\ell,0}\|_F + \frac{2\ell_T B_\ell M_{\ell-1}}{n}, \text{quad}B_{\ell-1} = s_{\ell-1} D_\ell B_\ell. \quad (82)$$

Projection contraction bounds the Variant A learned increment by the second term in D_ℓ ; the dense increment is no larger. Hence $\|\widehat{W}_\ell\|_F \leq D_\ell$, and the backward recurrence gives the next $B_{\ell-1}$. Finally,

$$\|W_1(t)\|_F \leq \|W_{1,0}\|_F + 2SB_1X. \quad (83)$$

All physical bounds are independent of P . At fixed order, the moment integrals are bounded because $1 \leq \tau \leq \ell_T$ and every history is bounded. Standard ODE continuation gives global physical-time existence for every P . Repeating (75) with D_ℓ in place of R gives (40) for all orders.

B Supplementary empirical tables

B.1 Full focused old/new-clock comparison

Table 4 reports the focused width-2048 panel at each model's own first training-RMS 0.05 endpoint. The maximum Euler step is 1/64, TF32 is disabled, and every displayed model reaches the training target.

For the adverse GELU alternating-quadrant case, halving the maximum step to 1/128 changes old-clock P1–P5 errors to 2.99053, .21233, .11999, .02364, .02150 and new-clock errors to 3.07505, 2.90895, .11200, .22258, .11200. This is an endpoint numerical sensitivity result, not a continuous-gradient-flow certificate.

Table 4: Full-circle RMS from a freshly fitted dense reference. “Old” is the activity clock and “New” is the response-arclength clock.

Activation/depth	Task	Clock	P1	P2	P3	P4	P5
tanh/2	Alternating outliers	Old	.07250	.04009	.01674	.00524	.00123
tanh/2	Alternating outliers	New	.07018	.04901	.01323	.00786	.00324
tanh/2	Alternating quadrant	Old	.05417	.06186	.02388	.00818	.00658
tanh/2	Alternating quadrant	New	.06552	.09918	.01283	.00591	.00346
tanh/2	Paired quadrant	Old	.00566	.00135	.00056	.00022	.00007
tanh/2	Paired quadrant	New	.00833	.00124	.00111	.00032	.00019
GELU/3	Alternating outliers	Old	.44504	.07570	.00549	.00207	.00196
GELU/3	Alternating outliers	New	.45472	.11902	.04500	.00679	.00098
GELU/3	Alternating quadrant	Old	2.98734	.20941	.11963	.01469	.01279
GELU/3	Alternating quadrant	New	3.06966	2.69207	1.24526	1.21472	.99568
GELU/3	Paired quadrant	Old	.08807	.00574	.00443	.00149	.00077
GELU/3	Paired quadrant	New	.11162	.00674	.00157	.00206	.00016

Table 5: Representative measured resources. “Total” adds the fixed initialized internal matrix or matrices to moving state and excludes solver workspace. CUDA peak includes implementation workspace.

Experiment	Model	Moving MiB	Moving+fixed MiB	CUDA peak MiB	Integration s
MNIST100, $n = 4096$	Dense	152.531	152.531	1941.881	102.976
	P1	30.781	158.781	544.266	139.893
	P2	37.031	165.031	603.141	149.922
	P3	43.281	171.281	654.391	152.269
Deep circle, $n = 4096$	Dense	256.094	256.094	2592.9–2849.0	task-dependent
	P1	1.094	257.094	432.063	slower than dense
	P2	2.094	258.094	432.063	slower than dense
	P3	3.094	259.094	432.063	slower than dense

B.2 Measured state and runtime examples

The measured peak reduction comes from the particular solver representation; the mathematical state ledger shows no total-state reduction once W_0 is counted. The closure’s durable advantage in the present form is that the learned moving correction and stored history are low rank and independent of elapsed training time.

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